Nasal epithelial gene expression improves the prediction of asthma exacerbations in the ATLANTIS study

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Supplementary material

Supplementary methods

ALLIANCE cohort

Clinical and nasal microarray expression data from an independent cohort, the All Age Asthma Cohort (ALLIANCE), was used for the replication of the findings. ALLIANCE is a prospective, multicenter observational study established to investigate asthma and wheezing disorders across the lifespan, from infancy to adulthood. Recruitment includes patients with asthma or recurrent wheeze, defined according to the Global Initiative for Asthma (GINA) 2014 guidelines, as well as age- and sex-matched healthy controls. Inclusion criteria required the ability to read and understand German and to provide informed consent, while individuals with a history of respiratory tract infection within the four weeks preceding enrolment were excluded. Participants undergo comprehensive phenotyping through standardized clinical assessments, structured interviews, validated questionnaires, and telephone follow-ups, complemented by extensive biomaterial collection including blood, nasal and pharyngeal swabs, nasal epithelial brushes, induced sputum, and exhaled breath condensates [14,15]. Lung function and airway inflammation are assessed regularly using harmonized protocols. The study is registered at ClinicalTrials.gov (NCT02419274).

From the ALLIANCE cohort, adult asthma patients were selected for the replication analysis if they had multiple available timepoints, with the second visit occurring within 24 months of the first, recorded exacerbation history, and transcriptomic data from their baseline visit. In total, 137 patients met these criteria and were included in the replication analysis. Exacerbations were defined as episodes requiring systemic corticosteroid treatment or hospitalization for at least three days [15].

A detailed description of the transcriptomic profiling was published previously [15]. Briefly, nasal epithelial samples were collected using sterile dental brushes (IDB-G50, Top Caredent) by brushing the medial and superior surfaces of the inferior nasal meatus. Brushes were transferred into precooled cryotubes containing RNAprotect Cell Reagent (Qiagen), cut with a heavy wire cutter, and stored at −80 °C. Total RNA was extracted using the AllPrep RNA Micro Kit (Qiagen), and RNA quantity and integrity were assessed with a NanoDrop spectrophotometer and the Agilent 2100 Bioanalyzer using the RNA 6000 Nano Chip Kit. RNA was amplified and Cy3-labeled using the Low Input Quick Amp Labeling Kit (Agilent Technologies) and hybridized to SurePrint G3 Human Gene Expression 8×60K Microarrays in a single batch. Data processing and quality control were performed with GeneSpring GX v14.9.1 (Agilent Technologies), excluding compromised signals and normalizing expression data using the 80th percentile shift method.

Supplementary tables

Supplementary Table 1. Significantly differentially expressed genes between follow-up exacerbators and non-exacerbators identified in the ATLANTIS study.

Gene	logFC	PValue	FDR	External gene name
ENSG00000234389	1.49	4.67e-07	0.00799	AC007278.1
ENSG00000213386	1.34	1.13e-06	0.00969	AC022217.1
ENSG00000251139	1.57	2.19e-06	0.00987	AC084871.1
ENSG00000279838	0.804	2.31e-06	0.00987	AL356273.3
ENSG00000259671	1.42	2.89e-06	0.0099	MTCYBP23
ENSG00000259600	1.45	4.57e-06	0.013	AC066616.2
ENSG00000287937	1.55	5.98e-06	0.0146	AC079753.2
ENSG00000281005	0.774	7.54e-06	0.0161	LINC00921
ENSG00000259379	1.38	9.76e-06	0.0186	MTND5P32
ENSG00000123610	1.37	1.24e-05	0.0211	TNFAIP6
ENSG00000178363	0.786	1.35e-05	0.0211	CALML3
ENSG00000110427	-1.11	1.68e-05	0.024	KIAA1549L
ENSG00000224609	0.914	1.91e-05	0.0252	HSD52

ENSG00000143207	0.197	2.36e-05	0.0287	COP1
ENSG00000276900	0.893	2.51e-05	0.0287	AC023157.2
ENSG00000286813	0.568	3.25e-05	0.0347	AC015871.6
ENSG00000115607	1.08	3.61e-05	0.0363	IL18RAP
ENSG00000157540	0.115	4.35e-05	0.0414	DYRK1A
ENSG00000250069	0.652	5.1e-05	0.0433	AC011379.1
ENSG00000146592	0.929	5.52e-05	0.0433	CREB5
ENSG00000261026	1.37	5.65e-05	0.0433	AC105046.1
ENSG00000173559	0.788	5.65e-05	0.0433	NABP1
ENSG00000163803	0.403	5.83e-05	0.0433	PLB1
ENSG00000140030	1.09	6.57e-05	0.0464	GPR65
ENSG00000273272	1.03	6.78e-05	0.0464	U62317.3
ENSG00000206531	1.2	7.44e-05	0.0489	CD200R1L

Model corrected for age, sex and smoking status. Significance level was defined as FDR < 0.05. logFC: log fold change, FDR: false discovery rate.

Supplementary Table 2. Baseline clinical characteristics of the ALLIANCE cohort participants.

Variable	ALLIANCE cohort (N=137)	Non-exacerbators (N=100)	Exacerbators (N=37)	p

Follow-up duration, n (%)				0.2
12 months	40 (29%)	32 (32%)	8 (22%)	
24 months	97 (71%)	68 (68%)	29 (78%)	
Male, n (%)	61 (45)	43 (43)	18 (49)	0.6
Age, years (median [IQR])	**55 [47-64]**	**54 [40-63]**	**58 [53-69]**	**0.001**
Smoking, n (%)				0.5
Never	99 (73)	71 (71)	28 (80)	
Ever	35 (26)	28 (28)	7 (20)	
NA	1	1	0	
BMI, kg/m2 (median [IQR])	25.4 [23.4-29.7]	24.9 [23.4-29.0]	26.5 [23.0-31.6]	0.14
Pack-years, (median [IQR])	6 [2-18]	5 [2-18]	6 [5-20]	0.5
NA	103	73	30	
ACQ-6, (median [IQR])	**1.00 [0.33-2.33]**	**0.67 [0.33-2.08]**	**1.50 [0.83-2.50]**	**0.013**
GINA, severe (4-5), n (%)	**46 (34)**	**24 (24)**	**22 (59)**	**<0.001**
Exacerbation last year, n (%)				**<0.001**
Yes	**45 (33)**	**22 (22)**	**23 (62)**	
No	**91 (66)**	**77 (77)**	**14 (38)**	

NA	1	1	0	
Pre-bronchodilator FEV1, % pred (mean (SD))	81 (21)	83 (19)	74 (26)	0.069
NA	1	0	1	
FVC, % pred (median [IQR])	104 [94-115]	105 [97-115]	100 [79-113]	0.059
Pre-bronchodilator FEV_1/FVC (median [IQR])	**66 [55-73]**	**67 [57-74]**	**58 [50-70]**	**0.018**
Pre-bronchodilator FEF50, % pred (median [IQR])	48 [23-79]	49 [29-81]	38 [16-69]	0.2
NA	70	43	27	
ICS usage, yes, n (%)				0.12
Yes	112 (82)	80 (80)	32 (86)	
No	23 (17)	20 (20)	3 (8.1)	
NA	2 (1.5)	0	2 (5.4)	
Systemic corticosteroids usage, yes, n (%)				**<0.001**
Yes	21 (15)	8 (8.0)	13 (35)	
No	116 (85)	92 (92)	24 (65)	
Biologics usage, yes, n (%)				0.2
Yes	29 (21)	24 (24)	5 (14)	
No	107 (78)	76 (76)	31 (84)	

NA	1 (0.7)	0	1 (2.7)	
RV/TLC, % of predicted (mean (SD))	**40 (10)**	**38 (9)**	**47 (10)**	**0.003**
NA	53	33	20	
Blood eosinophils, 10^9/L (median [IQR])	0.18 [0.09-0.36]	0.19 [0.10-0.35]	0.15 [0.06-0.47]	0.4
NA	1	0	1	
Blood neutrophils, 10^9/L (median [IQR])	**3.89 [3.08-5.00]**	**3.73 [3.01-4.59]**	**4.43 [3.23-5.95]**	**0.025**
NA	1	1	0	
Sputum eosinophils, % (median [IQR])	2.00 [1.00-4.00]	2.00 [2.00-4.00]	2.00 [1.00-5.00]	>0.9
NA	75	61	14	
Sputum neutrophils, % (median [IQR])	29 [-1-62]	41 [-1-62]	14 [-1-54]	0.5
NA	53	33	20	
Sputum lymphocytes, % (median [IQR])	**0.40 [-1.00-1.25]**	**0.50 [-1.00-1.30]**	**0.00 [-1.00-0.40]**	**0.038**
NA	53	33	20	
Sputum epithelial cells, % (median [IQR])	0.5 [-1.0-2.5]	0.6 [-1.0-2.5]	0.0 [-1.0-3.6]	0.3
NA	53	33	20	

Supplementary Table 3. Multivariate regression model for future exacerbations constructed using clinical variables and small airway measurement (R5-R20) in the ATLANTIS cohort.

	Clinical model + small airways	
Predictor	B (SE)	P
Sex (male)	-0.30 (0.38)	0.43
Smoking (never)	-0.098 (0.41)	0.81
ACQ-6 score	0.10 (0.17)	0.53
FEV_1 % of pred.	-0.03 (0.01)	0.003*
Exacerbation history (yes)	1.02 (0.45)	0.02*
Blood eosinophil counts 10^9/L	0.84 (0.80)	0.30
GINA (severe)	1.01 (0.40)	0.01*
R_{5-20}	$-1.19*10^{-5}$	0.87

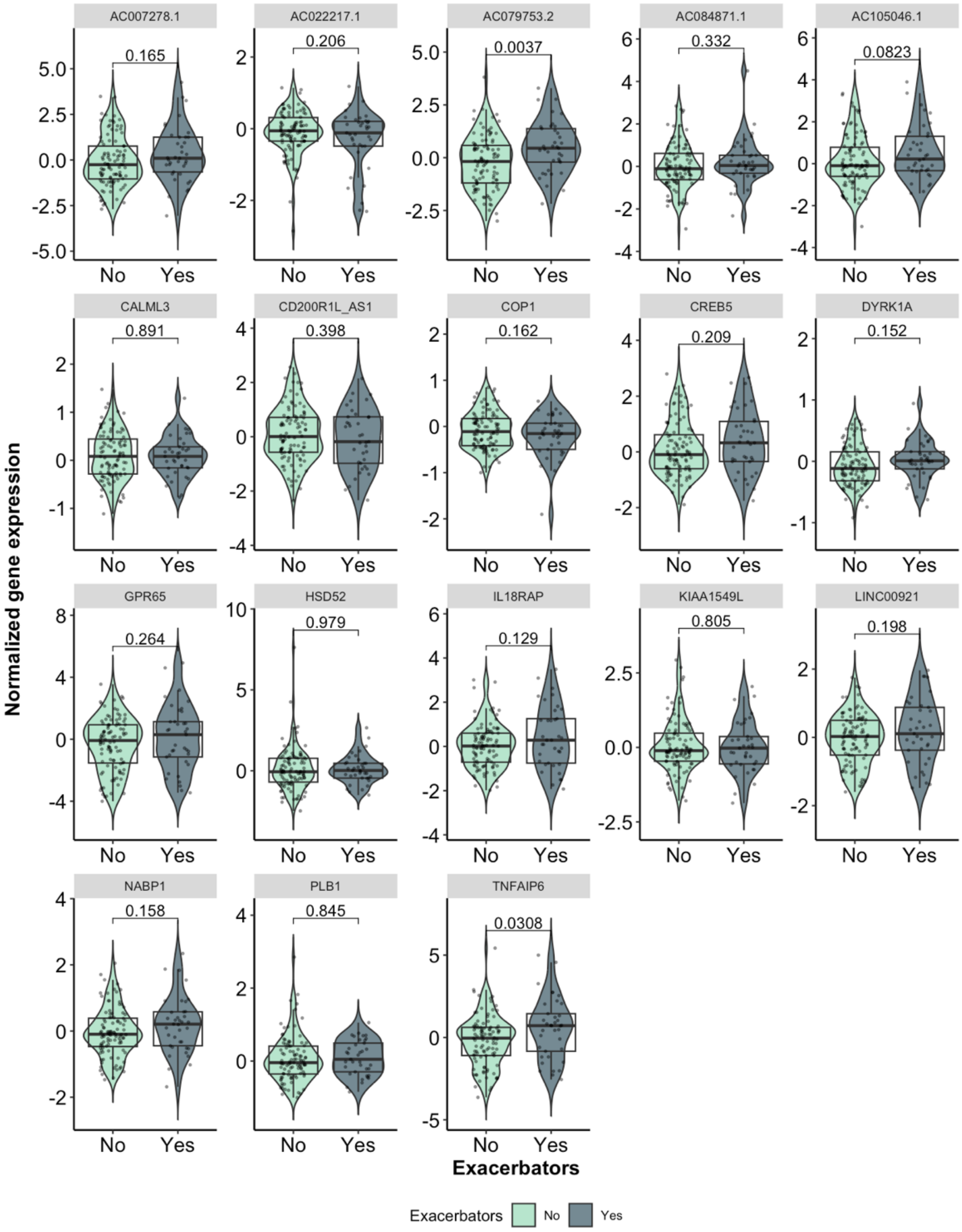
Description: ACQ-6: asthma control questionnaire 6; GINA: Global Initiative for Asthma stages: GINA (severe) was defined as GINA stages 4 and 5; FEV_1 % of pred.: Forced expiratory volume in 1 second percentage of predicted;R5-20: resistance at 5 Hz–resistance at 20 Hz; B: beta coefficient; SE: standard error; P: p-value.

Supplementary *Table 4. Performance of logistic regression models for asthma exacerbation prediction in the ATLANTIS (n=298) and ALLIANCE (n=131) cohorts, evaluated using repeated 5-fold cross-validation (500 repeats).*

MODEL	ATLANTIS			ALLIANCE		
	AUC	Bal. acc.	Sens.	AUC	Bal. acc.	Sens.
sex + smoking	0.55 [0.4–0.69]	0	0 [0-0]	0.52 [0.29-0.71]	0.5 [0.5-0.5]	0[0-0]
sex + smoking + ACQ-6	0.61 [0.45–0.75]	0.50 [0.48–0.5]	0 [0-0]	0.61 [0.38-0.82]	0.51 [0.45-0.58]	0.05 [0 – 0.29]
sex + smoking + ACQ-6 + FEV_1 % pred	0.65 [0.49–0.8]	0.53 [0.48–0.59]	0.07 [0–0.24]	0.6 [0.36- 0.83]	0.51 [0.45- 0.62]	0.06 [0 – 0.29]
sex + smoking + ACQ-6 + FEV_1 % pred + Exac. history	0.69 [0.53–0.84]	0.57 [0.48–0.66]	0.17 [0–0.33]	0.69 [0.46-0.89]	0.57 [0.42-0.72]	0.23 [0-0.5]
sex + smoking + ACQ-6 + FEV_1 % pred + Exac. history + Blood eosinophils	0.71 [0.55–0.86]	0.59 [0.49–0.7]	0.22 [0–0.42]	0.67 [0.44-0.88]	0.56 [0.42-0.71]	0.22 [0-0.5]
sex + smoking + ACQ-6 + FEV_1 % pred + Exac. history + Blood eosinophils + GINA	0.74 [0.58–0.87]	0.61 [0.5-0.73]	0.26 [0.08–0.5]	0.67 [0.44-0.87]	0.55 [0.42-0.7]	0.21 [0-0.5]
sex + smoking + ACQ-6 + FEV_1 % pred + Exac. history + Blood eosinophils + GINA + ssGSEA	0.75 [0.59-0.9]	0.63 [0.51–0.75]	0.30 [0.08-0.54]	0.67 [0.43-0.88]	0.58 [0.42-0.76]	0.28 [0-0.67]

Description: Models were constructed by sequential addition of clinical predictors, with the ssGSEA composite score added as the final step. Model performance is reported using the area under the curve (AUC), balanced accuracy (Bal. acc.), and sensitivity (Sens.), reported as median [95% CI] across cross-validation repeats. Smoking status was categorized as ever versus never smokers. Asthma Control Questionnaire-6 (ACQ-6) corresponds to the total score. Forced expiratory volume in one second percent predicted (FEV_1 % predicted) was used as a measure of lung function. Exacerbation history was categorized as yes versus no based on the occurrence of at least one exacerbation in the previous year. Blood eosinophil counts are expressed as cells x 10^9/L. Global Initiative for Asthma (GINA) severity was *classified as mild versus severe asthma. Single sample Gene Set Enrichment Analysis (ssGSEA) scores included the genes previously identified in ATLANTIS as differentially expressed between exacerbators and non-exacerbators.*

Supplementary figures



Supplementary Figure 1. Baseline expression values from the ALLIANCE cohort of the exacerbation-associated genes identified in the ATLANTIS cohort compared between follow-up exacerbators and non-exacerbators.